



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**SECOND YEAR FIRST SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF BUSINESS ADMINISTRATION WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

BIT 211: DATABASE SYSTEMS I

Date: 12th February, 2018

Time: 8.30 -11.30 am

INSTRUCTIONS:

- Answer question ONE and any other TWO questions



QUESTION ONE. (COMPULSORY 25 MARKS)

- a). Define Database Management Systems (DBMS) (2 marks)
- b) Explain the use of the following objects in databases (6 marks)
- i) Tables
 - ii) Forms
 - ii) Query
- c) The following car details are to be entered in a database: Make, Date-of-manufacture, colour, and cost.
- i) Prepare a suitable database structure showing field names and their field data types (6 mks)
 - ii) Explain how you index the data such that cars of the same make and color are together. (2mks)
 - iii) Write a statement that can be used to select cars of color green. (2mks)
 - iv) Give an instruction that would:
 - Select cars whose cost is between Kshs 500,000/= and Kshs 1,000,000/= (3mks)
 - Determine the average cost of all cars. (3mks)

QUESTION TWO. (15 MARKS)

- a) Describe four major shortcomings of the conventional (paper) file structures that are being addressed by the database approach. (8marks)
- b) Explain TWO methods used to secure data in a database (4 marks)

- c) Explain how 'parents' and 'child' tables are related in relational database (3 marks)

QUESTION THREE. (15 MARKS)

a) Differentiate between:

i) Primary key and relationship (2mks)

ii) Tabular and columnar form layout. (2mks)

b) State three types of relationships that can be used to link tables in databases. (3 marks)

c) i) Define the term normalization as used in database design (2 marks)

ii) State the three objectives normalization. (6 marks)

QUESTION FOUR. (15 MARKS)

a) Explain three types of database models. (6mks)

b) List four examples of database programs in the market today (4 marks)

c) State the components of a Database hierarchy in ascending order (5 marks)

QUESTION FIVE. (15 MARKS)

a) Explain **THREE** main disadvantages of database (6 marks)

b) Explain **THREE** different types of integrity constraint in database. (6 marks)

c) State **THREE** roles of database administrator (3 marks)



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MAIN CAMPUS

BIT 212: HARDWARE SUPPORT AND MAINTANANCE

Date: 13th February, 2018

Time: 12.00 - 3.00 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions.
- Question ONE carries 25 marks and the rest 15 marks each



QUESTION ONE (COMPULSORY 25 MARKS)

- a). Highlight **FIVE** basic safety precautions to use when working on a computer: (5 marks)
- b) State **FOUR** ways to help shield against power fluctuation and power problems (8 marks)
- c) Explain **TWO** ways by which processors are classified (4marks)
- d) Explain **FOUR** different types of RAM and ROM (8 marks)

QUESTION TWO (15 MARKS)

- a) Identify **FIVE** common hardware problems and how to solve them (10 marks)
- b) Explain **FIVE** eye safety precautions that has to be observed when using computers. (5 marks)

QUESTION THREE (15 MARKS)

- a) Using a well drawn diagram explains the basic organization of a computer system (9 marks)
- b) Explain different types of buses and how they work (6 marks)

QUESTION FOUR (15 MARKS)

- a) Identify **TWO** types of power unit and their functions (4mks)
- b) Describe **FOUR** functions performed by the CPU (8 marks)
- c). In reference to **ALU**, explain the meaning of Logic Operations and give an example of this processing operation (3 Marks)

QUESTION FIVE (15 MARKS)

- a) What is the computer Motherboard? (2 marks)
- b) State any **FIVE** components found on the computer motherboard (5 marks)
- c) By giving examples, differentiate between Primary and Secondary Computer Storage (8 marks)



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**SECOND YEAR FIRST SEMESTER EXAMINATION FOR
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INFORMATION TECHNOLOGY**

MAIN CAMPUS

BIT 213: INFORMATION RESOURCES MANAGEMENT

Date 28th February, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other THREE



QUESTION ONE (COMPULSORY) (25 marks)

- (a) Explain the meaning of the expression Information Resource Management. [3 marks]
- (b) State the benefits of Information Resources Management in an organization. [5 marks]
- (c) Describe the origins and evolution of Information Resources Management. [12 marks]
- (d) Elucidate on the most significant issues on the future success of Information Resources Management. [5 marks]

QUESTION TWO

- (a) Enumerate on the three disciplines of Information Resources Management. [6 marks]
- (b) Describe the three classes of Information Resources Management. [9 marks]

QUESTION THREE

“Online Information Publications is a strategic to any scholar or Institution.” In light of the foregoing statement, describe various examples of such online Information publications while pointing out the advantages.

[15 marks]

QUESTION FOUR

- (a) Describe Information related risks that contribute to enterprise risks. [7 marks]
- (b) Describe areas to focus on for an audit of Information Management. [8 marks]

QUESTION FIVE

- (a) Explain the concept of Information Management Strategic Planning Process. [7 marks]
- (b) “Strategic Planning determines the organizational readiness for Data Warehousing or Information Management Initiatives by comparing the current situation with laid down guidelines.” In light of the foregoing statement, list and explain such information management guidelines to consider. [8 marks]



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MAIN CAMPUS

BIT 214: INTRODUCTION TO COMPUTER PROGRAMMING

Date: 15th February, 2018

Time: 12.00 - 3.00 pm

INSTRUCTIONS:

- Answer question ONE and any other THREE questions.



QUESTION ONE. (COMPULSORY 25 MARKS)

- a). Explain the following terms as used in programming (4 marks)
- i. Object code
 - ii. Translators
- b) Explain **THREE** advantages of high level programming languages (6 marks)
- c) Maseno University savings society Pays **5%** interest on shares exceeding **100,000** shillings and **3%** on shares that do not meet this target. However, no interest is paid on deposits in the member's MSS bank account.

Required: Design a pseudo code for a program that would:

- i) Prompt the user for shares and deposit of particular members
 - ii) Calculate the interest and total savings
 - iii) Display the interest and total savings on the screen for particular members of the society. (7mks)
- d) Draw a flowchart for the above pseudo code. (8mks)

QUESTION TWO. (15 MARKS)

- a) Briefly explain the stages involved in programme development. Highlight main activities in each stage. (12 marks)
- b) Explain any **THREE** selection control structure. (3 marks)

QUESTION THREE. (15 MARKS)

- a) Describe **THREE** features of fourth generation languages (4GLS) (6mks)
- b) List **THREE** examples of fourth generation programming languages. (3mks)
- c) Describe **THREE** factors to be considered while choosing a programming language (6marks)

Question Four. (15 marks)

- a) Draw a flowchart showing (3mks)
 - i. The nested IF selection
 - ii. The FOR loop
 - iii. REPEAT.... UNTIL loop

- b) Write HTML code that would start a web page with title '**MANAGEMENT SCIENCE**'.

Headline 1 SBE, and headline 2 **EXAM SECOND YEAR**. Display a green background with the marquee "**HELLO**" appearing on the screen.

(6marks)

- c) State the guidelines for designing good pseudo codes (6 marks)

QUESTION FIVE. (15 MARKS)

- a) Give two differences between the interpreters and compilers (2 marks)
- b) Write a pseudo code that would be used to classify animals according to sex. If a letter M is input the program should display "male" otherwise it should display "female" (4 marks)

- c) The gross salary of employees in LISUMU BOOK ENTERPRISE is based on basic salary and additional benefits as follows. Employees who have worked for the company for more than 10 years receive an additional pay of 10% to their basic salary. Monthly salary bonus based on

monthly sales of books is as follows: above 500,000 monthly sales bonus rate is 15%, between 250,000 and 500,000 is 10% and below 250,000 monthly sales is 5%. Draw a flowchart for a program that would be used to calculate the gross salary then output basic, gross and all benefit. (9 marks)